Attracting invariant curves in strongly convex heavy ball dynamics

Abstract

We construct fixed smooth strongly convex potentials on $\mathbb{R}^2$ whose heavy ball dynamics undergo supercritical Neimark–Sacker bifurcations. The first example has momentum $1/2$ and a two-cycle at step 1. The second has momentum $9/10$, global Hessian bounds $I/101 \preceq \nabla^2 f_3 \preceq I$, and a prime three-cycle at step $7/2$. Its bifurcation lies strictly inside the stability region for every quadratic with the same bounds, with margin $3/10$ at the critical step. For the one fixed potential f_3, varying the step produces three disjoint attracting invariant curves cyclically permuted by the physical map. We give the complete global potential, derive its local jets from the defining minimization, and calculate the actual moving-cycle crossing and a strictly negative cubic coefficient exactly. The classical invariant-curve theorem supplies local attraction in the full four-dimensional state space. Attracting finite cycles and the bifurcation theorem are prior results; the construction verifies their nonlinear compatibility within a fixed globally admissible potential. No irrational-rotation or equal-position initialization claim is made.

Keywords: heavy ball dynamics; Neimark–Sacker bifurcation; invariant curves; quadratic stability; strongly convex potential

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1 Two realizations and quadratic stability

For $0 < \beta < 1$ and a fixed smooth potential $V : \mathbb{R}^2 \to \mathbb{R}$, consider

$$x_{n+1} = (1 + \beta)x_n - \gamma \nabla V(x_n) - \beta x_{n-1}. \tag{1}$$

Its state is the pair of consecutive positions. All quadratics with Hessian spectrum in $[\mu, L]$, where $0 < \mu \leq L$, are strictly stable whenever $0 < \gamma L < 2(1 + \beta)$. This is the exact uniform stability range for the class, as follows from the characteristic polynomial $z^2 - (1 + \beta - \gamma t)z + \beta$ for $\mu \leq t \leq L$.

Nonlinear attracting cycles within this region are already known. Lessard, Recht and Packard [7, Appendix B] give an attracting scalar three-cycle with $\mu = 1$, $L = 25$, $\beta = 4/9$ and $\gamma = 1/9$; their equal-position initialization reaches its basin. Here $\gamma L = 25/9 < 26/9$. Goujaud, Taylor and Dieuleveut [2, Theorems 4.2 and 5.2] establish attracting higher-period cycles and C^∞ smoothing inside the quadratic-stable region. Their Figures 4 and 7 use $\beta = .75$, $L = 1$ and $\gamma = 3.3 < 3.5$. These results already establish smooth strongly convex nonlinear cycling under quadratically stable parameters.

The question studied here concerns a further local change of the invariant set: can one fixed globally smooth strongly convex potential undergo a nondegenerate birth of

attracting curves from a periodic orbit as only the step varies? We give two explicit realizations. The three-cycle example also satisfies the strict global quadratic-stability restriction. Its existence is the stronger conclusion below.

Theorem 1. *The function f_3 defined by (82) is C^∞ on $\mathbb{R}^2$ and satisfies*

$$\frac{1}{101} I \preceq \nabla^2 f_3(x) \preceq I \qquad (x \in \mathbb{R}^2). \tag{2}$$

It has a unique minimizer x_*. Set $\beta = 9/10$, $\gamma_* = 7/2$, and

$$\mathcal{F}_\gamma(x, v) = \left(\frac{19}{10} x - \gamma \nabla f_3(x) - \frac{9}{10} v, x \right), \qquad \mathcal{G}_\gamma = \mathcal{F}_\gamma^3. \tag{3}$$

Let $w_i = (\cos(2\pi i/3), \sin(2\pi i/3))^T$, with indices modulo three. The states $W_i = (w_i, w_{i-1})$ form a prime three-cycle at γ_*. They continue smoothly as a three-cycle of the same fixed objective. At the critical parameter the return spectrum is

$$\lambda,\ \bar{\lambda},\ \kappa\lambda,\ \kappa\bar{\lambda}, \qquad \lambda = \frac{50 + i\sqrt{2986941}}{1729}, \quad \kappa = \frac{729}{1000}. \tag{4}$$

Along the actual moving branch,

$$\frac{\mathrm{d}}{\mathrm{d}\gamma} |\lambda(\gamma)| \Big|_{\gamma_*} = \frac{4130846306178349}{14519314142774106} > 0. \tag{5}$$

The cubic radial coefficient, in the normalization of Section 9, is strictly negative.

There exists $0 < \varepsilon_0 < 3/20$ such that, for every $\gamma_* < \gamma < \gamma_* + \varepsilon_0$, the map $\mathcal{G}_\gamma$ has three disjoint attracting embedded C^1 circles $\Gamma_i(\gamma)$ near W_i. They are cyclically permuted by $\mathcal{F}_\gamma$, and their union is locally asymptotically stable in $\mathbb{R}^4$, with an open local basin. They shrink to the three cycle states as $\gamma \downarrow \gamma_*$. Their current-position projections satisfy

$$\|\nabla f_3(x)\| \geq \frac{1}{2}, \qquad \|x - x_*\| \geq \frac{1}{2}. \tag{6}$$

The continued cycle is locally asymptotically stable just below γ_* and unstable just above it. The entire enclosing interval $|\gamma - \gamma_*| < 3/20$ lies within $0 < \gamma L < 2(1 + \beta)$ for the same global constant $L = 1$.

The local width ε_0 is existential; the exact margin at the central step is $19/5 - 7/2 = 3/10$. The invariant set consists of three one-dimensional curves in a four-dimensional phase space. The assertion allows phase locking and does not require irrational rotation at every parameter. Nor does it assert that the equal-position diagonal meets the basin, or that every point near the unstable cycle approaches the curves.

The period matters for the strict step restriction. If $p \neq q$ form a two-cycle of (1), subtraction of its two update equations gives

$$\gamma\big(\nabla V(p) - \nabla V(q)\big) = 2(1 + \beta)(p - q). \tag{7}$$

Thus a globally L-Lipschitz gradient forces $\gamma L \geq 2(1+\beta)$. No refinement of a valid global Lipschitz estimate can place a two-cycle strictly below this boundary.

Sections 2–6 present the separate two-cycle realization, with its own potential f, step η, and momentum $1/2$. All notation and statements within those sections refer to that example. Sections 7–10 construct and prove Theorem 1, using the different potential f_3 and momentum $9/10$. Neither potential is varied with its step. The smooth maximum, Moreau regularization and classical Neimark–Sacker theorem are standard tools; the work is to verify the compatible global bounds, moving orbit and nonlinear coefficient for the displayed systems.

2 The discrete system and the result

For a smooth potential $f : \mathbb{R}^2 \to \mathbb{R}$, consider the heavy ball recurrence

$$w_{n+1} = \frac{3}{2}w_n - \eta\nabla f(w_n) - \frac{1}{2}w_{n-1}, \qquad \eta > 0. \tag{8}$$

The state is the ordered pair of current and previous positions:

$$F_\eta(w, v) = \left(\frac{3}{2}w - \eta\nabla f(w) - \frac{1}{2}v, w\right), \qquad G_\eta = F_\eta^2. \tag{9}$$

The objective remains fixed while η varies. A simple closed invariant curve means an embedded copy of S^1 in the four-dimensional state space; it need not be a geometric circle. Attraction below is local asymptotic stability as a compact invariant set, including Lyapunov stability and convergence of distance from a neighborhood.

Oscillatory momentum trajectories have long been studied, including the learning examples of Rojas [1]. Smooth strongly convex cycling examples and their robustness are established by Goujaud, Taylor and Dieuleveut [2]. Regis [3, Section 5] gives the heavy ball two-cycle spectral pencil and explicitly distinguishes the information supplied by the determinant and inertia from later complex stability exits. The question addressed here is a concrete nonlinear realization: a globally regular convex potential with a nondegenerate birth of attracting invariant curves from a two-cycle.

The linear pencil and the Neimark–Sacker theorem are existing machinery. Our task is to verify, for one fixed potential, the global realization, actual moving-cycle crossing, and nonlinear coefficient to which that theorem applies. The construction and all decisive calculations are explicit. No claim of complete priority over the older momentum-learning or dissipative-map literature is made.

Let $e = (1, 0)^T$. Section 3 defines the single function f used throughout.

Theorem 2. *The function f defined in (23) is C^∞ on $\mathbb{R}^2$ and satisfies*

$$\frac{3}{4}I \preceq \nabla^2 f(w) \preceq 100I \qquad (w \in \mathbb{R}^2). \tag{10}$$

In particular it has a unique minimizer w_*. For the actual map (9)*, the states*

$$Z_0 = (-e, e), \qquad Z_1 = (e, -e) \tag{11}$$

form a two-cycle at $\eta = 1$. This cycle has a smooth local continuation $Z_0(\eta), Z_1(\eta)$.

At $\eta = 1$, the spectrum of $DG_1(Z_0)$ is

$$\frac{-4+3i}{5}, \quad \frac{-4-3i}{5}, \quad \frac{-4+3i}{20}, \quad \frac{-4-3i}{20}. \tag{12}$$

If $\lambda(\eta)$ denotes the critical multiplier on the moving branch with $\lambda(1) = (-4+3i)/5$, then

$$\frac{\mathrm{d}}{\mathrm{d}\eta}|\lambda(\eta)|\Big|_{\eta=1} = \frac{30}{41}, \qquad d_{\mathrm{NS}} = -\frac{3\sqrt{5}}{8200}. \tag{13}$$

The coefficient d_{NS} uses the eigenvector and Taylor convention in Section 5.

There is $\delta > 0$ such that, for every $1 < \eta < 1 + \delta$, the return map G_η has two disjoint attracting C^1 simple closed invariant curves $\Gamma_0(\eta), \Gamma_1(\eta)$, with

$$F_\eta \Gamma_0(\eta) = \Gamma_1(\eta), \qquad F_\eta \Gamma_1(\eta) = \Gamma_0(\eta). \tag{14}$$

Their union is a locally asymptotically stable invariant set for F_η. The curves shrink to Z_0, Z_1, respectively, as $\eta \downarrow 1$. After decreasing δ, every position w in their current-position projections satisfies

$$\|\nabla f(w)\| \geq 1, \qquad \|w - w_*\| \geq \frac{1}{100}. \tag{15}$$

At the critical step the prescribed initialization is $w_{-1} = e$, $w_0 = -e$. We do not assert that equal initial positions reach this cycle or its neighboring invariant sets. The bifurcation is supercritical: the cycle is linearly asymptotically stable for $\eta < 1$ sufficiently close to 1 and loses stability as the critical pair moves outward. The theorem concerns invariant curves and set attraction, without requiring irrational rotation for every parameter.

3 A fixed global potential

All constants in this section are independent of η. Put

$$\tau = \frac{1}{100}, \qquad \kappa = \frac{1}{1000}, \qquad r = \frac{3\sqrt{5}}{10}, \qquad R_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{16}$$

Define the positive definite matrices

$$H_i = \frac{3}{2} I - r R_i, \quad K_i = H_i (I - \tau H_i)^{-1}, \quad D_i = I + \tau K_i = (I - \tau H_i)^{-1}, \qquad i = 0, 1, \tag{17}$$

and the points and gradients

$$p_0 = -\frac{97}{100} e, \quad p_1 = \frac{97}{100} e, \qquad g_0 = -3e, \quad g_1 = 3e. \tag{18}$$

For $\zeta_i = z - p_i$, let

$$Q_0(z) = g_0^T \zeta_0 + \frac{1}{2}\zeta_0^T K_0 \zeta_0 + \frac{\kappa}{4}(e^T D_0 \zeta_0)^4, \tag{19}$$

$$Q_1(z) = g_1^T \zeta_1 + \frac{1}{2}\zeta_1^T K_1 \zeta_1. \tag{20}$$

Use the normalized even bump

$$\rho(u) = \begin{cases} c\exp[-1/(1-u^2)], & |u| < 1, \\ 0, & |u| \geq 1, \end{cases} \qquad c^{-1} = \int_{-1}^{1} \exp[-1/(1-u^2)]\,\mathrm{d}u, \tag{21}$$

and define

$$\sigma(s) = \int_{-1}^{1} |s-u|\rho(u)\,\mathrm{d}u, \qquad \Phi(z) = \frac{Q_0(z) + Q_1(z) + \sigma(Q_0(z) - Q_1(z))}{2}. \tag{22}$$

The desired potential is the quadratic infimal convolution

$$f(w) = \min_{z \in \mathbb{R}^2} \left\{ \Phi(z) + \frac{\|w - z\|^2}{2\tau} \right\}. \tag{23}$$

This is the classical Moreau regularization [4, Section 7]. The following proof establishes its global smoothness and exact local coefficients directly for the specified function.

3.1 Global convexity and exact patch selection

Let $h_\pm = 3/2 \pm r$ and $m = h_-/(1 - \tau h_-)$. Since $r < 7/10$, both H_i have eigenvalues in $(4/5, 11/5)$. Consequently

$$K_i \succeq mI, \qquad \|K_i\| < \frac{1100}{489} < \frac{9}{4}, \qquad \|D_i\| < \frac{500}{489} < \frac{21}{20}. \tag{24}$$

The quartic in Q_0 is convex. More explicitly, with $d_0 = D_0 e$,

$$\nabla^2 Q_0(z) = K_0 + 3\kappa(d_0^T \zeta_0)^2 d_0 d_0^T \succeq mI, \qquad \nabla^2 Q_1 = K_1 \succeq mI. \tag{25}$$

The bump is C^∞ on the entire line. Differentiating the convolution gives

$$\sigma'(s) = 2\int_{-\infty}^{s} \rho(u)\,\mathrm{d}u - 1, \qquad \sigma''(s) = 2\rho(s). \tag{26}$$

Thus σ is C^∞, $|\sigma'| \leq 1$, and $\sigma'' \geq 0$. Evenness gives $\int u\rho(u)\,\mathrm{d}u = 0$, so $\sigma(s) = |s|$ for $|s| \geq 1$.

Writing $\Delta = Q_0 - Q_1$ and $a = (1 + \sigma'(\Delta))/2 \in [0, 1]$, we obtain

$$\nabla^2 \Phi = a\nabla^2 Q_0 + (1-a)\nabla^2 Q_1 + \frac{1}{2}\sigma''(\Delta)\nabla\Delta\nabla\Delta^T \succeq mI. \tag{27}$$

This verifies strong convexity throughout the transition region as well as within the two polynomial pieces.

At its own center $Q_i(p_i) = 0$. Let $d = 97/50 < 2$, so $p_1 - p_0 = de$. The competing values obey

$$Q_1(p_0) = -3d + \frac{d^2}{2} e^T K_1 e < -\frac{291}{50} + \frac{9}{2} = -\frac{33}{25} < -1, \tag{28}$$

$$\begin{aligned} Q_0(p_1) &= -3d + \frac{d^2}{2} e^T K_0 e + \frac{\kappa}{4} (de^T D_0 e)^4 \\ &< -\frac{33}{25} + \frac{1}{4000} \left(\frac{21}{10}\right)^4 < -1. \end{aligned} \tag{29}$$

The positive term in the last line is $194481/40000000 < 1/100$. By continuity, $Q_i - Q_{1-i} > 1$ on a neighborhood of p_i. Hence the smooth maximum satisfies the exact local identity

$$\Phi = Q_i \quad \text{on a neighborhood of } p_i. \tag{30}$$

There is no approximation error in this selection.

3.2 The global envelope and its derivatives

For each w, the expression minimized in (23) is coercive and $(m+1/\tau)$-strongly convex in z. It therefore has a unique minimizer $P(w)$, characterized by

$$w = T(z) := z + \tau \nabla \Phi(z), \qquad z = P(w). \tag{31}$$

Existence and uniqueness make T a bijection. Its derivative $I + \tau \nabla^2 \Phi$ is positive definite everywhere. The inverse function theorem gives smooth local inverses, and uniqueness makes them the global C^∞ inverse P. Substitution in (23) and differentiation give

$$\nabla f(w) = \frac{w - P(w)}{\tau} = \nabla \Phi(P(w)), \qquad \nabla^2 f(w) = A(I + \tau A)^{-1}, \quad A = \nabla^2 \Phi(P(w)). \tag{32}$$

An eigenvalue $a \geq m$ of A becomes $a/(1 + \tau a)$. It follows that

$$h_- I = \frac{m}{1 + \tau m} I \preceq \nabla^2 f(w) \preceq 100 I. \tag{33}$$

Since $h_- > 4/5 > 3/4$, this proves (10). Integrating the Hessian along a line segment proves the global gradient bound. Strong convexity makes f coercive, so its minimizer exists and is unique.

3.3 Local Taylor expansions of the actual envelope

Set $w_0 = -e$, $w_1 = e$. By (18), $w_i = p_i + \tau g_i$. Therefore (31) gives $P(w_i) = p_i$, and

$$\nabla f(w_i) = g_i, \qquad \nabla^2 f(w_i) = K_i D_i^{-1} = H_i. \tag{34}$$

Continuity of P allows the use of the exact local pieces (30). Near w_1, write $w = w_1 + \xi$ and $P(w) = p_1 + \zeta$. The proximal equation reduces to $\xi = D_1 \zeta$, giving

$$\nabla f(w_1 + \xi) = g_1 + H_1 \xi. \tag{35}$$

Thus f is exactly quadratic up to affine and constant terms near w_1.

Near w_0, put $P(w_0+\xi)=p_0+\zeta$ and $t=e^TD_0\zeta$. The exact equation is

$$\xi=D_0\zeta+\tau\kappa t^3D_0e,\qquad \zeta=D_0^{-1}\xi-\tau\kappa t^3e. \tag{36}$$

Multiplication by e^TD_0 gives

$$t+\tau\kappa(e^TD_0e)t^3=\xi_1. \tag{37}$$

Its local inverse is smooth and odd, with $t=\xi_1-\tau\kappa(e^TD_0e)\xi_1^3+O(|\xi_1|^5)$. Since $D_0e-\tau K_0e=e$, the gradient identity is

$$\begin{aligned}\nabla f(w_0+\xi)&=g_0+H_0\xi+\kappa t^3e\\&=g_0+H_0\xi+\kappa\xi_1^3e+O(\|\xi\|^5).\end{aligned} \tag{38}$$

In particular,

$$D^3f(w_0)=D^3f(w_1)=0. \tag{39}$$

The cubic force in (38) is consequently a derived jet of the globally defined envelope.

4 The cycle and the moving spectral crossing

The inverse map is explicit:

$$F_\eta^{-1}(u,w)=\big(w,3w-2\eta\nabla f(w)-2u\big). \tag{40}$$

Thus F_η is a global C^∞ diffeomorphism. From (34), direct substitution gives

$$F_1(-e,e)=(e,-e),\qquad F_1(e,-e)=(-e,e). \tag{41}$$

These are transitions of the actual fixed potential, not only of its linearization.

4.1 The return spectrum

Write $A_i=3I/2-H_i=rR_i$. Then

$$J_i=DF_1(Z_i)=\begin{pmatrix}A_i & -I/2\\ I & 0\end{pmatrix},\qquad M=DG_1(Z_0)=J_1J_0=\begin{pmatrix}A_1A_0-I/2 & -A_1/2\\ A_0 & -I/2\end{pmatrix}. \tag{42}$$

For later reproduction, the full matrix is

$$M=\begin{pmatrix}-1/2 & -9/20 & 0 & -3\sqrt{5}/20\\ 9/20 & -1/2 & -3\sqrt{5}/20 & 0\\ 3\sqrt{5}/10 & 0 & -1/2 & 0\\ 0 & -3\sqrt{5}/10 & 0 & -1/2\end{pmatrix}. \tag{43}$$

Block elimination, initially for $\lambda\neq-1/2$ and then by polynomial identity, yields the two-cycle pencil

$$\det(\lambda I-M)=\det\big((\lambda+1/2)^2I-\lambda A_1A_0\big). \tag{44}$$

This is the pencil appearing in Regis [3, Section 5.1]. Here $A_1A_0 = r^2\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$, and therefore

$$\begin{aligned}\chi(\lambda) := \det(\lambda I - M) &= (\lambda + 1/2)^4 + r^4\lambda^2 \\ &= \left(\lambda^2 + \frac{8}{5}\lambda + 1\right)\left(\lambda^2 + \frac{2}{5}\lambda + \frac{1}{16}\right). \end{aligned} \tag{45}$$

This proves (12). Put $\lambda_0 = (-4+3i)/5$. Its relevant powers are

$$\lambda_0^2 = \frac{7-24i}{25}, \quad \lambda_0^3 = \frac{44+117i}{125}, \quad \lambda_0^4 = \frac{-527-336i}{625}. \tag{46}$$

Each has nonzero imaginary part. Thus $\lambda_0^j \neq 1$ for $j = 1, 2, 3, 4$, and all four eigenvalues in (12) are distinct.

4.2 The branch motion is included

Since $1 \notin \operatorname{spec} M$, the implicit function theorem applied to $G_\eta(Z) - Z = 0$ gives a smooth branch $Z_0(\eta) = (a(\eta), b(\eta))$ through Z_0. The second component of this equation forces

$$F_\eta(a(\eta), b(\eta)) = (b(\eta), a(\eta)) =: Z_1(\eta). \tag{47}$$

The two partners remain distinct nearby, since $a(1) = -e$ and $b(1) = e$. They satisfy

$$\eta\nabla f(a) = \frac{3}{2}(a-b), \qquad \eta\nabla f(b) = \frac{3}{2}(b-a). \tag{48}$$

For example, their derivatives are

$$a'(1) = \frac{1}{13}\begin{pmatrix}-25-\sqrt{5}\\ -5+5\sqrt{5}\end{pmatrix}, \qquad b'(1) = \frac{1}{13}\begin{pmatrix}25-5\sqrt{5}\\ -5+\sqrt{5}\end{pmatrix}. \tag{49}$$

These follow by solving the differentiated equations $-rR_0a' + 3b'/2 = 3e$ and $3a'/2 - rR_1b' = -3e$.

Let $\mathcal{H}_0(\eta) = \nabla^2 f(a(\eta))$ and $\mathcal{H}_1(\eta) = \nabla^2 f(b(\eta))$. The chain rule and (39) give

$$\mathcal{H}_0'(1) = D^3 f(w_0)[a'(1), \cdot, \cdot] = 0, \qquad \mathcal{H}_1'(1) = 0. \tag{50}$$

Consequently the first derivative of the actual site matrix $3I/2 - \eta\mathcal{H}_i(\eta)$ equals $-H_i$ at 1. This justifies using the fixed matrices H_i for this first derivative; the cycle itself has not been frozen.

For a concise calculation set

$$u = \frac{3}{2}(1-\eta), \qquad v = \eta r, \qquad \widehat{A}_i(\eta) = uI + vR_i. \tag{51}$$

The trace of $\widehat{A}_1\widehat{A}_0$ is $2u^2$ and its determinant is $(u^2-v^2)^2$. Its corresponding characteristic polynomial is

$$\widehat{\chi}(\lambda, \eta) = (\lambda + 1/2)^4 - 2u^2\lambda(\lambda + 1/2)^2 + (u^2 - v^2)^2\lambda^2. \tag{52}$$

By (50), the actual characteristic polynomial on the moving branch has the same first parameter derivative at $\eta = 1$. In particular

$$\partial_\eta \widehat{\chi}(\lambda, 1) = 4r^4\lambda^2 = \frac{81}{100}\lambda^2. \tag{53}$$

The critical root is simple. Implicit differentiation, using (45), gives

$$\lambda'(1) = -\frac{(81/100)\lambda_0^2}{\chi'(\lambda_0)} = \frac{-48 + 186i}{205}, \qquad \bar{\lambda}_0\lambda'(1) = \frac{30 - 24i}{41}. \tag{54}$$

Since $|\lambda_0| = 1$, taking real parts proves the crossing derivative in (13). The complementary pair remains strictly inside the unit disk. Hence all four multipliers are inside for $\eta < 1$ sufficiently close to 1, whereas the critical pair is outside for $\eta > 1$ sufficiently close to 1.

5 The nonlinear coefficient and the invariant curves

The coefficient must be computed for the return map at the actual cycle. Its exact quadratic and cubic terms are particularly simple because the second local force is affine.

5.1 The full return-map tensor

For a displacement $X = (x, y) \in \mathbb{R}^2 \times \mathbb{R}^2$ about Z_0, the first step has current-position displacement

$$u = A_0 x - \frac{1}{2}y - \kappa x_1^3 e + O(\|X\|^5). \tag{55}$$

For small enough X, its image lies in the exact quadratic neighborhood of w_1. Composing the second step therefore gives

$$G_1(Z_0 + X) - Z_0 = MX - \kappa x_1^3 V + O(\|X\|^5), \qquad V = \begin{pmatrix} A_1 e \\ e \end{pmatrix} = (0, r, 1, 0)^T. \tag{56}$$

Use the Taylor convention

$$MX + \frac{1}{2}B(X, X) + \frac{1}{6}C(X, X, X) + o(\|X\|^3). \tag{57}$$

Equation (56) gives the full real tensors, extended complex multilinearly when necessary:

$$B = 0, \qquad C(U, W, Y) = -6\kappa U_1 W_1 Y_1 V. \tag{58}$$

The first coordinate here is the first current-position coordinate.

5.2 Normalization and exact coefficient

Choose the right eigenvector with first coordinate 1:

$$q = \left(1, -i, \frac{-1 - 2i}{\sqrt{5}}, \frac{2 - i}{\sqrt{5}}\right)^T. \tag{59}$$

The corresponding normalized left row is

$$\ell = \left(\frac{14-3i}{41}, \frac{3+14i}{41}, \frac{\sqrt{5}(-5+4i)}{82}, \frac{\sqrt{5}(4+5i)}{82} \right). \tag{60}$$

Direct multiplication yields

$$Mq = \lambda_0 q, \quad \ell M = \lambda_0 \ell, \quad \ell q = 1, \quad \ell \bar{q} = 0. \tag{61}$$

In the usual Hermitian convention $p = \bar{\ell}^T$, so $M^T p = \bar{\lambda}_0 p$ and $\bar{p}^T q = 1$. The contractions needed below are

$$\ell V = \frac{\sqrt{5}(-8+31i)}{205}, \qquad \bar{\lambda}_0 \ell V = \frac{\sqrt{5}(5-4i)}{41}. \tag{62}$$

Since $q_1 = \bar{q}_1 = 1$, the tensor in (58) gives $C(q, q, \bar{q}) = -6\kappa V$.

In the Neimark–Sacker convention of Kuznetsov and Sacker [6], the vanishing of B removes both quadratic correction terms. The resonant cubic coefficient and its radial real part are

$$c_{21} = \frac{1}{2} \ell C(q, q, \bar{q}) = -3\kappa \ell V, \qquad d_{\mathrm{NS}} = \mathrm{Re}(\bar{\lambda}_0 c_{21}). \tag{63}$$

The factor can also be checked directly. On the linear center space write $X = zq + \bar{z}\bar{q}$, so $x_1 = z + \bar{z}$. Projection by ℓ makes the coefficient of $z^2\bar{z}$ in the cubic term equal to $-3\kappa\ell V$. The full quadratic tensor is zero, so the quadratic center-manifold graph has zero forcing; its homological denominators are nonzero by the strict separation of the stable multipliers from unit-modulus products. A cubic stable graph contributes nothing through the linear projection ℓM. The nonresonant cubic coordinate changes do not alter the resonant coefficient. Thus (63) is the coefficient of the actual reduced map, with no uncomputed quadratic correction.

Substitution of (62) gives

$$d_{\mathrm{NS}} = -3\kappa \frac{5\sqrt{5}}{41} = -\frac{3\sqrt{5}}{8200} < 0. \tag{64}$$

It is essential to take the real part after multiplication by $\bar{\lambda}_0$. The numerical value depends on the eigenvector scale: $\|q\|^2 = 4$, and the unit vector $q/2$, with left row 2ℓ, gives $d_{\mathrm{NS}}/4 = -3\sqrt{5}/32800$. The sign is unchanged.

5.3 Application of the classical map theorem

We apply the finite-dimensional invariant-curve theorem in Sacker [5, Chapter II, Sections 3–4, Theorems 1–2]. It concerns a smooth map with a simple unit pair and a strictly contracting complementary block. A positive derivative of the log multiplier modulus, absence of the strong third and fourth resonances, and a negative real cubic coefficient in exponential normal form give an asymptotically stable invariant closed curve

for all sufficiently small positive parameters. Choosing regularity index 2 gives a C^1 curve. The theorem includes attraction in the ambient space through its stable complementary block.

Here set $\mu = \eta - 1$ and translate by $Z_0(1+\mu)$. This is a jointly C^∞ family with the origin fixed, and all derivatives required by the theorem are bounded on a sufficiently small compact neighborhood. The critical eigenvalues are the simple nonreal pair $\lambda_0, \bar{\lambda}_0$. Equation (46) checks the strong resonances. The complementary pair is $\lambda_0/4, \bar{\lambda}_0/4$. A real basis formed from its eigenvectors puts this block into a rotation-dilation of Euclidean operator norm $1/4$, verifying the source's contraction condition. Smooth local choices of these bases are available because the eigenvalues are simple and separated.

The linear growth quantity in that theorem is

$$A = \frac{\mathrm{d}}{\mathrm{d}\mu} \log |\lambda(1+\mu)| \Big|_{\mu=0} = \frac{30}{41} > 0. \tag{65}$$

Its cubic damping quantity is

$$B = \mathrm{Re}(c_{21}/\lambda_0) = d_{\mathrm{NS}} < 0. \tag{66}$$

The notation A, B in (65)–(66) denotes scalar normal-form quantities, not the earlier site matrices or Taylor tensor. Positive coordinate rescalings in the normal-form theorem preserve both signs. All hypotheses of the map theorem therefore hold.

It follows that G_η has an attracting C^1 simple closed invariant curve $\Gamma_0(\eta)$ near $Z_0(\eta)$ for every $1 < \eta < 1+\delta$, after choosing a sufficiently small $\delta > 0$. It shrinks to Z_0 as $\eta \downarrow 1$. For orientation, in the specified complex normal-form coordinate its leading radius is

$$|z| = \sqrt{-A/B}\sqrt{\mu} + O(\mu) = 20\,5^{1/4}\sqrt{\mu} + O(\mu). \tag{67}$$

This is not an assertion of a Euclidean radius in the original four-dimensional coordinates. The leading polar term and its C^1 correction give a simple closed curve, and the theorem supplies attraction in all four state dimensions. It does not require irrational rotation numbers for every nearby parameter; phase locking is compatible with the conclusion.

5.4 The physical one-step dynamics

Define $\Gamma_1(\eta) = F_\eta \Gamma_0(\eta)$. Since F_η is a diffeomorphism, its image is again a C^1 simple closed curve. The commutation of F_η and $G_\eta = F_\eta^2$ yields

$$G_\eta \Gamma_1 = F_\eta G_\eta \Gamma_0 = \Gamma_1, \qquad F_\eta \Gamma_1 = G_\eta \Gamma_0 = \Gamma_0. \tag{68}$$

The second curve shrinks to Z_1, so the two curves lie in disjoint neighborhoods for small enough δ. Conjugacy by F_η transfers local asymptotic stability for G_η from Γ_0 to Γ_1.

Their union is invariant under the actual map F_η. Even iterates of a state sufficiently near either component approach that component under G_η; continuity on a compact local

neighborhood makes odd iterates approach the other component. Lyapunov stability transfers as well: for a neighborhood of the union, first choose a smaller neighborhood whose image under F_η lies inside it, and then use G_η-stability for the even iterates. This proves local asymptotic stability of the union for F_η.

Finally, $\|\nabla f(w_i)\| = 3$ and the position projections shrink to w_i. Decrease δ so these projections lie within $1/50$ of the corresponding sites. The global Lipschitz bound gives

$$\|\nabla f(w)\| \geq 3 - 100\|w - w_i\| \geq 1. \tag{69}$$

Since $\nabla f(w_*) = 0$, it also gives $\|w - w_*\| \geq 1/100$. This completes the proof of Theorem 2.

6 Interpretation and reproducibility

The example shows a local transition from a stable two-cycle to an attracting union of two curves for a fixed globally strongly convex potential. The one-step invariant set has two one-dimensional components. A two-dimensional torus would arise only in a different construction, such as a continuous-time suspension. No statement about a global basin or equal-position initialization is used.

The step lies outside the range that stabilizes every scalar curvature up to the global Lipschitz bound. Indeed any nonconstant heavy ball two-cycle a, b, with momentum β and gradient Lipschitz constant Λ, satisfies

$$\eta\big(\nabla f(a) - \nabla f(b)\big) = 2(1 + \beta)(a - b). \tag{70}$$

Taking norms gives $\eta\Lambda \geq 2(1 + \beta)$. Thus the present construction makes no claim of instability within a universal quadratic stability range. It is a discrete dynamical example, not a worst-case complexity lower bound.

The dimension and the noncommuting local Hessians have a concrete role. In one position dimension, the determinant of a two-step heavy ball derivative is $\beta^2 < 1$, so both return multipliers cannot form a unit conjugate pair. If the two symmetric Hessians in higher dimension commute, they can be simultaneously diagonalized. Each scalar return factor then has determinant β^2, and its nonreal pair has modulus $\beta < 1$. The matrices in (17) avoid this commuting situation.

The smooth maximum and Moreau envelope are standard convex constructions. The source comparison is correspondingly narrow. The older momentum-learning literature already contains oscillatory examples, the smooth strongly convex cycling mechanism is established in [2], and the linear two-cycle pencil is explicit in [3]. The invariant-curve existence theorem and coefficient convention are classical [5, 6]. Some older dissipative-map sources were available only through abstracts in the accompanying literature check, so no exhaustive priority conclusion is drawn. The result established here is the displayed globally regular potential and the exact nonlinear bifurcation verified for its actual recurrence.

The supplementary file `check_coefficients.py` verifies the characteristic factorization, eigenvector normalizations, branch derivatives, transverse crossing, and cubic coefficient using exact arithmetic over $\mathbb{Q}(\sqrt{5}, i)$. Those checks supplement the analytic global-convexity and invariant-curve arguments; numerical iteration is not used as proof. Formula (23) specifies a mathematical potential. One update of (8) calls its gradient once; no claim about the cost of evaluating that envelope or solving its defining minimization is made.

7 The global three-cycle potential

We now define the second potential, denoted by f_3. All constants in this section are fixed independently of the varying step γ. Put

$$\beta = \frac{9}{10}, \quad \gamma_* = \frac{7}{2}, \quad \kappa = \frac{729}{1000}, \quad \tau = \frac{1}{10}, \quad P = \frac{117}{50}, \quad Q = -\frac{9}{10}, \tag{71}$$

and let $h > 0$ be determined by

$$h^2 = \frac{(1-\kappa)^2 + \kappa\tau^2/(1+\kappa)^2}{-PQ} = \frac{226837536481}{6295762746000}. \tag{72}$$

Define

$$A_0 = \begin{pmatrix} 9/5 & 0 \\ 0 & 3/2 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 9/5 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 167/117 & h \\ h & -3/2 \end{pmatrix}, \qquad H_i = \frac{(19/10)I - A_i}{7/2}. \tag{73}$$

The positions w_i are those in Theorem 1. With $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, set

$$g_i = \frac{57I - \sqrt{3}J}{70} w_i, \qquad y_i = w_i - g_i = \frac{13I + \sqrt{3}J}{70} w_i, \qquad b_i = -\frac{813}{2450}, \tag{74}$$
$$a = \frac{1}{100}, \qquad \ell = (1,1)^T, \qquad (e_0, e_1, e_2) = (0,0,1).$$

The local matrix data will be realized by a convex cutoff before applying the smooth maximum. Define

$$K_i = H_i(I - H_i)^{-1}, \quad B_i = K_i - aI, \quad v_i = (I - H_i)^{-1}\ell, \quad R_i(s) = \sqrt{s^T B_i s + 2e_i(v_i^T s)^4}. \tag{75}$$

Lemma 1. *All quantities in* (75) *are real and well-defined. They satisfy*

$$\frac{9}{50} < h < \frac{1}{5}, \qquad \frac{1}{35} I \preceq H_i \prec I, \qquad B_i \succeq \frac{33}{1700} I. \tag{76}$$

For $i \neq j$, the support gaps

$$\Delta_{ij} = b_i - b_j - g_j^T(y_i - y_j) - \frac{a}{2}\|y_i - y_j\|^2 \tag{77}$$

take precisely the two values $49971/245000$ *and* $60471/245000$.

Proof. The bounds on h follow by rational comparison in (72). The matrices simplify to

$$H_0 = \operatorname{diag}(1/35, 4/35), \quad H_1 = \operatorname{diag}(1/35, 19/35), \quad H_2 = \begin{pmatrix} 79/585 & -2h/7 \\ -2h/7 & 34/35 \end{pmatrix}.$$

The first diagonal entries of $H_2 - I/35$ and $I - H_2$ are positive, and their determinants are, respectively,

$$\frac{7515336342959}{77123093638500}, \qquad \frac{1679110467479}{77123093638500}.$$

Sylvester's criterion proves the Hessian bounds. Spectral calculus gives $K_i \succeq I/34$, and hence the asserted bound for B_i.

For the gaps, write $U = (13I + \sqrt{3}J)/70$ and $V = (57I - \sqrt{3}J)/70$. Then

$$U^T U = \frac{43}{1225} I, \qquad V^T U = \frac{369}{2450} I + \frac{\sqrt{3}}{70} J.$$

Distinct vertices have $w_j^T w_i = -1/2$, and $w_j^T J w_i$ is either $\sqrt{3}/2$ or $-\sqrt{3}/2$. Thus

$$-g_j^T (y_i - y_j) \in \left\{ \frac{1002}{4900}, \frac{1212}{4900} \right\}, \qquad \|y_i - y_j\|^2 = \frac{129}{1225}.$$

Subtracting $129/245000$ proves the two values in the statement. The smaller occurs for $(i, j) = (0, 1), (1, 2), (2, 0)$. ∎

Define the standard flat bump $b(t) = \exp(-1/t)$ for $t > 0$, and $b(t) = 0$ for $t \le 0$. This function is distinct from the constants b_i. Put

$$\chi(t) = \frac{b(2 - t)}{b(2 - t) + b(t - 1)} \quad (t \ge 0), \qquad \psi_\rho(t) = \int_0^t (t - s)\chi(s/\rho)\, \mathrm{d}s. \tag{78}$$

The positive constants used here are

$$\Delta = \frac{49971}{245000}, \quad \mathfrak{M} = \max_{i \ne j} R_j(y_i - y_j), \quad \rho = \frac{\Delta}{16(1 + \mathfrak{M})}, \quad \delta = \frac{\Delta}{16}. \tag{79}$$

The maximum is over six finite nonzero values, so this is an explicit definition of fixed positive numbers. Set

$$q_i(y) = b_i + g_i^T (y - y_i) + \frac{a}{2}\|y - y_i\|^2 + \psi_\rho(R_i(y - y_i)). \tag{80}$$

Let $\zeta(t) = c \exp[-1/(1 - t^2)]$ for $|t| < 1$ and zero otherwise, where c normalizes its integral to one. Define

$$q(y) = \int_{[-1,1]^3} \max_{0 \le i \le 2} \{q_i(y) + \delta t_i\} \prod_{i=0}^{2} \zeta(t_i)\, \mathrm{d}t_0 \mathrm{d}t_1 \mathrm{d}t_2, \tag{81}$$

$$f_3(x) = \min_{y \in \mathbb{R}^2} \left\{ q(y) + \frac{1}{2}\|x - y\|^2 \right\}. \tag{82}$$

These formulas specify the complete global potential. The cutoff is needed because extending all prescribed Hessians by unrestricted quadratic polynomials would not by itself preserve the support inequalities at the other nodes.

Proposition 7.1. The potential (82) is globally C^∞ and satisfies (2). At each of the three positions,

$$f_3(w_i) = 0, \quad \nabla f_3(w_i) = g_i, \quad \nabla^2 f_3(w_i) = H_i, \quad D^3 f_3(w_i) = 0, \quad D^4 f_3(w_i) = 24 e_i \ell^{\otimes 4}. \tag{83}$$

Proof. We first check the cutoff and gluing globally. The denominator in (78) is positive everywhere on $[0, \infty)$. The function χ is smooth, takes values in $[0, 1]$, equals one on $[0, 1]$ and vanishes on $[2, \infty)$. Consequently

$$\psi_\rho'' \geq 0, \quad \psi_\rho' \geq 0, \quad \psi_\rho(t) = \frac{t^2}{2} \ (0 \leq t \leq \rho), \quad 0 \leq \psi_\rho(t) \leq 2\rho t. \tag{84}$$

Each R_i is convex. Indeed, it is the Euclidean norm of the vector

$$\big(B_i^{1/2} s, \sqrt{2e_i}(v_i^T s)^2\big).$$

The final component is nonnegative and convex. Its convexity inequality, monotonicity of the norm in that nonnegative component, and the triangle inequality give the claimed convexity of R_i. Thus $\psi_\rho \circ R_i$ is convex. It is smooth away from zero; near zero it equals the polynomial $R_i^2/2$, so it is smooth there too. In particular, each q_i is globally smooth and a-strongly convex. Near its own center it is exactly

$$q_i(y_i + s) = b_i + g_i^T s + \frac{1}{2} s^T K_i s + e_i (v_i^T s)^4. \tag{85}$$

The integral in (81), regarded as a function of the three input values, is a convolution of $\max_i u_i$ with a compactly supported smooth product kernel. It is smooth, convex, coordinatewise nondecreasing, and translation equivariant in the direction $(1, 1, 1)$. Since each $q_i(y) - a\|y\|^2/2$ is convex, these properties show that $q(y) - a\|y\|^2/2$ is convex. Hence q is globally smooth and $\nabla^2 q \succeq aI$.

The branch selected near y_i is exact. For $j \neq i$, Lemma 1 and (84) give

$$q_i(y_i) - q_j(y_i) \geq \Delta - 2\rho R_j(y_i - y_j) \geq \Delta - \frac{\Delta \mathfrak{M}}{8(1 + \mathfrak{M})} > \frac{7}{8}\Delta > 2\delta. \tag{86}$$

This strict inequality persists on a neighborhood. For every point in the kernel support, the same i-th branch wins there. Evenness of ζ makes its additive mean zero. Therefore $q = q_i$ on that neighborhood, with no smoothing error in (85).

For each x, the expression minimized in (82) is coercive and $(1 + a)$-strongly convex. Its unique minimizer $Y(x)$ solves

$$x = Y(x) + \nabla q(Y(x)). \tag{87}$$

The derivative $I + \nabla^2 q$ is positive definite everywhere. Existence, uniqueness and the inverse function theorem give a global smooth inverse Y. Differentiating the minimum value yields

$$\nabla f_3(x) = x - Y(x), \qquad \nabla^2 f_3(x) = D(I + D)^{-1}, \quad D = \nabla^2 q(Y(x)). \tag{88}$$

The eigenvalues of D are at least a, so $a/(1+a) = 1/101$ is a global lower Hessian bound and 1 is a global upper bound. Strong convexity gives coercivity and the unique minimizer.

At a node, $w_i = y_i + g_i$ and $q = q_i$ imply $Y(w_i) = y_i$. Thus $\nabla f_3(w_i) = g_i$, and $f_3(w_i) = b_i + \|g_i\|^2/2 = 0$, because $\|g_i\|^2 = 813/1225$. The Hessian formula gives $\nabla^2 f_3(w_i) = K_i(I + K_i)^{-1} = H_i$.

To verify the higher derivatives from the actual potential, write $x = w_i + z$ and $Y(x) = y_i + s$. The local proximal equation is

$$z = (I + K_i)s + 4e_i(v_i^T s)^3 v_i. \tag{89}$$

Let $C_i = (I + K_i)^{-1} = I - H_i$. The equation is odd in (z, s), and $C_i v_i = \ell$. Its inverse expansion is consequently

$$s = C_i z - 4e_i \ell(\ell^T z)^3 + O(\|z\|^5).$$

Subtracting this from z in (88) gives

$$\nabla f_3(w_i + z) = g_i + H_i z + 4e_i \ell(\ell^T z)^3 + O(\|z\|^5). \tag{90}$$

This proves all the derivatives in (83). In particular, at nodes 0 and 1 the potential is exactly quadratic locally. ∎

8 The three-cycle and its spectral crossing

The equilateral identities

$$w_{i+1} = -\frac{1}{2}w_i + \frac{\sqrt{3}}{2}Jw_i, \qquad w_{i-1} = -\frac{1}{2}w_i - \frac{\sqrt{3}}{2}Jw_i$$

give

$$\frac{(1+\beta)w_i - \beta w_{i-1} - w_{i+1}}{\gamma_*} = \frac{57I - \sqrt{3}J}{70}w_i = g_i. \tag{91}$$

Proposition 7.1 therefore verifies the cycle in the literal recurrence. Since the three current positions are distinct, its least period is three.

Proposition 8.1. The return derivative at W_0 has spectrum (4). The cycle has a smooth continuation for the fixed objective f_3, and its critical multiplier has the actual moving-branch derivative (5).

Proof. Define $L(A) = \left(\begin{smallmatrix} A & -\beta I \\ I & 0 \end{smallmatrix}\right)$ and $L_i = L(A_i)$. Direct differentiation and multiplication give

$$M_* = D\mathcal{G}_{\gamma_*}(W_0) = L_2 L_1 L_0 = \begin{pmatrix} 43/25 & -9h/10 & -1953/1300 & 0 \\ 117h/50 & 0 & -81h/50 & 81/100 \\ 117/50 & 0 & -81/50 & 0 \\ 0 & -9/10 & 0 & 0 \end{pmatrix}. \tag{92}$$

The characteristic polynomial is

$$\begin{aligned}\mathfrak{p}(z) &= (z^2 - \tau z + \kappa)(z^2 + \kappa) - PQh^2 z^2 \\ &= z^4 - \tau z^3 + Sz^2 - \kappa\tau z + \kappa^2, \qquad S = 2\kappa - PQh^2. \end{aligned} \tag{93}$$

The defining equation for h is precisely

$$S = 1 + \kappa^2 + \frac{\kappa\tau^2}{(1+\kappa)^2}.$$

Thus, with $c = \tau/[2(1+\kappa)] = 50/1729$,

$$\mathfrak{p}(z) = (z^2 - 2cz + 1)(z^2 - 2\kappa cz + \kappa^2). \tag{94}$$

This gives exactly the four stated eigenvalues. They are simple, and the complementary pair has modulus $\kappa < 1$. Since $0 < c < 1/2$ and $c \neq 0$, the critical root satisfies $\lambda^j \neq 1$ for $j = 1, 2, 3, 4$. In particular, 1 is not in the spectrum.

Apply the implicit-function theorem to $\mathcal{G}_\gamma(W) - W = 0$. It gives a smooth local fixed-point branch $W_0(\gamma)$, and set $W_i(\gamma) = \mathcal{F}_\gamma^i W_0(\gamma)$ for $i = 1, 2$. The three states remain distinct in disjoint neighborhoods and therefore remain a prime three-cycle. If $x_i(\gamma)$ is the current position of $W_i(\gamma)$, the corresponding upper-left derivative block is

$$A_i(\gamma) = (1+\beta)I - \gamma\nabla^2 f_3(x_i(\gamma)).$$

The orbit is allowed to move. Its position contribution to the first matrix derivative vanishes for the proved reason

$$A_i'(\gamma_*) = -H_i - \gamma_* D^3 f_3(w_i)[x_i'(\gamma_*)] = -H_i. \tag{95}$$

For example, if $b_i^F = (-g_i, 0)^T$, the actual branch velocity is determined by

$$(I - M_*)W_0' = L_2 L_1 b_0^F + L_2 b_1^F + b_2^F.$$

Nothing requires this velocity to be zero.

We calculate the radial derivative using the reciprocal spectral structure. For $\Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, symmetry of every $A_i(\gamma)$ yields

$$L_i(\gamma)^T \Omega L_i(\gamma) = \beta\Omega, \qquad M(\gamma)^T \Omega M(\gamma) = \kappa\Omega. \tag{96}$$

The nearby quartet can consequently be written as

$$re^{i\theta}, \quad re^{-i\theta}, \quad (\kappa/r)e^{i\theta}, \quad (\kappa/r)e^{-i\theta},$$

where $r = |\lambda(\gamma)|$. Let $T = \operatorname{tr} M$ and $S = ((\operatorname{tr} M)^2 - \operatorname{tr}(M^2))/2$. With $c(\gamma) = \cos\theta(\gamma)$,

$$T = 2c(r + \kappa/r), \qquad S = r^2 + \kappa^2/r^2 + 4\kappa c^2.$$

At $r = 1$, differentiation and elimination of c' give

$$\Xi r' = S' - \frac{4\kappa c}{1+\kappa} T', \qquad \Xi = 2(1-\kappa)\left((1+\kappa) - \frac{4\kappa c^2}{1+\kappa}\right) > 0. \tag{97}$$

The sign is immediate from $(1+\kappa)^2 - 4\kappa c^2 = (1-\kappa)^2 + 4\kappa(1-c^2) > 0$.

In the actual step direction (95), put $\dot{L}_i = \left(\begin{smallmatrix} -H_i & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then

$$M' = \dot{L}_2 L_1 L_0 + L_2 \dot{L}_1 L_0 + L_2 L_1 \dot{L}_0. \tag{98}$$

Using (92) in $T' = \operatorname{tr} M'$ and $S' = TT' - \operatorname{tr}(MM')$ gives the exact values

$$\begin{aligned} T' &= \frac{103547}{45500}, & S' &= \frac{9(2346h^2+209)}{7000}, \\ \Xi &= \frac{2419885690462351}{2584371744500000}, & S' - \frac{4\kappa c}{1+\kappa} T' &= \frac{217412963483071}{816117393000000}. \end{aligned} \tag{99}$$

Their quotient in (97) is the positive rational number in (5). This is the derivative along the moving branch of the one fixed potential, as required. ∎

9 The actual cubic coefficient at the three-cycle

We use the Taylor convention

$$\mathcal{G}_{\gamma_*}(W_0 + z) - W_0 = M_* z + \frac{1}{2}\mathcal{B}(z,z) + \frac{1}{6}\mathcal{C}(z,z,z) + O(\|z\|^4). \tag{100}$$

Let q be a right eigenvector for λ, and let q_2 be the current-position component of $L_1 L_0 q$. In this section, q denotes this eigenvector; the scalar function used to define the envelope remains the one in (81). Normalize

$$(q_2)_1 = 1, \qquad p^* M_* = \lambda p^*, \qquad p^* q = 1. \tag{101}$$

The first normalization fixes the right-vector amplitude. The pairing $p^* q = 1$ alone would not do so.

Proposition 9.1. The normalization (101) is well-defined, $\mathcal{B} = 0$, and the cubic radial coefficient is

$$d_3 = \frac{1}{2}\operatorname{Re}\big(\bar{\lambda} p^* \mathcal{C}(q, q, \bar{q})\big) \tag{102}$$

$$= 12\left(\frac{18}{13} - \frac{5}{117h}\right)\frac{-9(699529194h - 26611205)}{427063000\,\Xi} < 0, \tag{103}$$

where Ξ is given in (97).

Proof. First we determine the right normalization from the recurrence. If u_i are the successive position components of a λ-eigenvector, then $u_3 = \lambda u_0$ and $u_2 = \lambda u_{-1}$. Eliminating u_0, u_1 in the three linear updates gives

$$\begin{pmatrix} \lambda^2 - \tau\lambda + \kappa & -Ph\lambda \\ -Qh\lambda & \lambda^2 + \kappa \end{pmatrix} u_2 = 0. \tag{104}$$

Here $A_1A_0 - \beta I = \operatorname{diag}(P, Q)$ is invertible. If the first component of u_2 were zero, the first row of (104) would force $u_2 = 0$, and then all components of the eigenvector would vanish. Thus the requested normalization is possible. Put

$$\nu = \frac{Qh\lambda}{\lambda^2 + \kappa}, \qquad q = \left(\frac{\lambda + 81/50}{P\lambda}, \frac{\nu}{Q}, \frac{1}{\lambda}, \frac{\nu}{\lambda} \right)^T. \tag{105}$$

The denominator $\lambda^2 + \kappa$ is nonzero because $|\lambda^2| = 1 > \kappa$. Substitution in (92) proves $M_* q = \lambda q$ and $q_2 = (1, \nu)^T$.

Using $|\lambda| = 1$ and $\operatorname{Re}\lambda = c$, one obtains

$$|\lambda^2 + \kappa|^2 = (1 - \kappa)^2 + 4\kappa c^2 = -PQh^2, \qquad |\nu|^2 = \frac{5}{13}, \quad \operatorname{Re}\nu = -\frac{5}{234h}. \tag{106}$$

In particular, if $s = \ell^T q_2 = 1 + \nu$, then

$$|s|^2 = \frac{18}{13} - \frac{5}{117h} > 0. \tag{107}$$

For a reproducible left normalization, let $k^T = (P, 0, -81/50, 0)$, so that $k^T q = 1$. The spectral projector for the simple eigenvalue gives

$$p^* = \frac{k^T \operatorname{adj}(\lambda I - M_*)}{\mathfrak{p}'(\lambda)}. \tag{108}$$

It satisfies all pairings in (101), and also $p^* \bar{q} = 0$.

We next compute the nonlinear tensor from the realized potential. Equation (83) makes every second derivative $D^2\mathcal{F}_{\gamma_*}(W_i)$ zero, and hence $\mathcal{B} = 0$. Only node 2 contributes a third derivative. If $u(\xi)$ is the current-position component of $L_1L_0\xi$, the composition rule gives

$$\mathcal{C}(\xi, \eta, \zeta) = -24\gamma_* \begin{pmatrix} \ell \\ 0 \end{pmatrix} (\ell^T u(\xi))(\ell^T u(\eta))(\ell^T u(\zeta)). \tag{109}$$

There are no quadratic cross terms. Therefore

$$\mathcal{C}(q, q, \bar{q}) = -24\gamma_* \begin{pmatrix} \ell \\ 0 \end{pmatrix} s|s|^2. \tag{110}$$

For clarity about the normal-form factor, in the real center representation $zq + \bar{z}\bar{q}$ the resonant monomial $z^2\bar{z}$ occurs three times in $\mathcal{C}/6$. Its projected coefficient is thus $p^*\mathcal{C}(q, q, \bar{q})/2$. Since $\mathcal{B} = 0$, there are no quadratic normal-form corrections. The quadratic center-manifold graph also vanishes: its homological equations compare unit-modulus

products of center multipliers with strictly contracting stable multipliers. Higher center-graph corrections have zero left projection. This gives exactly (102), in agreement with the standard formula [8, Section 2.1, equation (2.11)]. The classical construction of the normal form is not a new ingredient here.

It remains to evaluate the left contraction with an exact sign. For this calculation only, introduce the auxiliary matrix family

$$\mathcal{M}(t) = L(A_2 + t\ell\ell^T)L_1L_0.$$

This is an algebraic means of evaluating the tensor contraction, not a varying family of global objectives. Its derivative is

$$N = \mathcal{M}'(0) = \begin{pmatrix} P & Q & -81/50 & 0 \\ P & Q & -81/50 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad Nq = \begin{pmatrix} \ell \\ 0 \end{pmatrix} s. \tag{111}$$

Simple-eigenvalue differentiation gives $\lambda_t' = p^* N q$. Hence, if r_t' denotes its radial derivative,

$$d_3 = -12\gamma_*|s|^2 r_t', \qquad r_t' = \operatorname{Re}\left(\bar{\lambda} p^* \begin{pmatrix} \ell \\ 0 \end{pmatrix} s\right). \tag{112}$$

The matrices in $\mathcal{M}(t)$ retain the conformal identity (96). Thus (97) applies, now with the auxiliary derivatives

$$T_t' = \operatorname{tr} N = \frac{36}{25}, \qquad S_t' = T \operatorname{tr} N - \operatorname{tr}(M_* N) = \frac{9}{500}(234h - 5).$$

Substitution gives

$$\Xi r_t' = S_t' - \frac{4\kappa c}{1+\kappa} T_t' = \frac{9(699529194h - 26611205)}{1494720500}. \tag{113}$$

Combining (107), (112) and (113), with $\gamma_* = 7/2$, proves the exact expression (103). Finally, $h > 9/50$ makes both $18/13 - 5/(117h)$ and $699529194h - 26611205$ strictly positive by rational comparison, while $\Xi > 0$. The coefficient is therefore strictly negative.

∎

10 The three-curve invariant set

Proof of Theorem 1. Proposition 7.1 proves the global properties of the one fixed potential. Equation (91) proves the exact prime cycle. Proposition 8.1 establishes its smooth moving continuation, the simple unit pair and stable complement, all nonresonances through order four, and the outward crossing. Proposition 9.1 gives the strictly negative cubic radial coefficient for the actual return map.

Translate $\mathcal{G}_\gamma$ by $W_0(\gamma)$ and use $\varepsilon = \gamma - \gamma_*$. This gives a C^∞ family fixing the origin. Smooth real spectral coordinates separate the center and stable planes. At the critical

parameter the stable block is similar, in a real basis formed from a complex eigenvector, to κ times a rotation. Its norm in that basis is strictly below one. All hypotheses of the classical invariant-curve theorem [5, Chapter II, Sections 3–4] are therefore satisfied. In the theorem's exponential convention the real cubic coefficient is exactly $d_3 = \mathrm{Re}(c_{21}/\lambda)$; the factor one half has already been accounted for in (102). The regularity here exceeds the five derivatives needed for its C^1 conclusion.

For every sufficiently small $\varepsilon > 0$, that theorem gives an embedded attracting C^1 curve $\Gamma_0(\gamma)$ of the return map. The attraction is in the full ambient variables: the theorem controls both the radial deviation and the strictly contracting complementary coordinates. Its positive leading center radius is proportional to $\sqrt{\varepsilon}$, with a small differentiable correction, so the curve shrinks to W_0 and is a simple closed curve. The conclusion is an invariant curve for each sufficiently small positive parameter, with no Diophantine premise.

The one-step map is a global smooth diffeomorphism, since

$$\mathcal{F}_\gamma^{-1}(X, x) = \left(x, \frac{(1+\beta)x - \gamma \nabla f_3(x) - X}{\beta} \right). \tag{114}$$

Define $\Gamma_1 = \mathcal{F}_\gamma \Gamma_0$ and $\Gamma_2 = \mathcal{F}_\gamma^2 \Gamma_0$. They are embedded C^1 circles, and shrinking the parameter interval puts them in disjoint neighborhoods of the distinct cycle states. Since $\mathcal{G}_\gamma \Gamma_0 = \Gamma_0$,

$$\mathcal{F}_\gamma \Gamma_0 = \Gamma_1, \qquad \mathcal{F}_\gamma \Gamma_1 = \Gamma_2, \qquad \mathcal{F}_\gamma \Gamma_2 = \Gamma_0. \tag{115}$$

The other two curves are attracting for the return map by conjugacy.

Let U_0 be a sufficiently small open attracting neighborhood of Γ_0 for $\mathcal{G}_\gamma$. If $z \in U_0$, then $\mathrm{dist}(\mathcal{G}_\gamma^n z, \Gamma_0) \to 0$. Uniform continuity near the compact curve of each of the finitely many maps $\mathcal{F}_\gamma^j$, $j = 0, 1, 2$, gives

$$\mathrm{dist}(\mathcal{F}_\gamma^{3n+j} z, \Gamma_j) \longrightarrow 0.$$

The same argument applies on the open neighborhoods $\mathcal{F}_\gamma^i U_0$. Return-map Lyapunov stability and continuity of the two intermediate iterates prove Lyapunov stability of the union: choose a smaller return-map neighborhood whose intermediate images lie within the prescribed neighborhood of the union. This proves local asymptotic stability as a compact invariant set in $\mathbb{R}^4$.

The continued cycle is stable just below the critical step because its critical radius is $1 + r'\varepsilon + o(\varepsilon) < 1$, while the complement remains stable. It is unstable just above because that radius exceeds one. Thus a ball containing the unstable cycle state cannot be entirely contained in the basin of the surrounding curves.

Finally, $\|g_i\| = \sqrt{813}/35 > 1/2$. By continuity and uniform shrinking of the three curves, every current-position projection satisfies the first inequality in (6) after a common reduction of the parameter interval. The global unit Lipschitz bound and $\nabla f_3(x_*) = 0$ give

$$\|\nabla f_3(x)\| \le \|x - x_*\|,$$

which proves its second inequality. Choose $\varepsilon_0 < 3/20$ smaller than every interval used above. Throughout $|\gamma - 7/2| < 3/20$,

$$0 < \frac{67}{20} < \gamma < \frac{73}{20} < \frac{19}{5} = 2(1+\beta). \tag{116}$$

For each quadratic eigenvalue $t \in [1/101, 1]$, the scalar characteristic polynomial in Section 1 satisfies the three strict Schur inequalities

$$1 - \beta > 0, \qquad \gamma t > 0, \qquad 2(1+\beta) - \gamma t > 0.$$

Thus every such quadratic is strictly stable on this same window. This completes the proof. ∎

The global upper Hessian bound, rather than the separate Hessians along the orbit alone, is essential to this comparison. The cutoff, smooth maximum and envelope prove that the displayed potential belongs to the claimed class at every position. The two-cycle secant identity (7) also shows that the stronger conclusion is not an improved constant for the earlier example: the two potentials and their momenta are different.

The source comparison remains limited. Theorem 4.2 of [2] makes perturbations of its canonical finite cycle evolve as heavy ball on $\mu\|x\|^2/2$. Its local Hessians are μI; its smoothing lemma preserves the existing locally affine gradients. Those statements already give attracting smooth higher-period cycles under strict quadratic stability. They do not prescribe the Hessians and quartic jets used here or establish a step bifurcation for one fixed potential. The scalar attracting cycle in [7] supplies an additional earlier strict-region example, and the two-cycle pencil in [3] concerns the linear return structure. Our finite-dimensional calculation verifies a particular simultaneous global realization and nonlinear bifurcation; it is not a general classification of heavy ball dynamics or a new invariant-curve theorem.

For [2], theorem-level comparison uses the open arXiv version of 9 October 2025. The publisher record identifies the journal publication, but its typeset body was not available in the source check. Some other recent and older leads were not available as complete primary texts. Consequently no exhaustive priority assertion is made. The classical map theorem and coefficient convention are credited explicitly to [5, 6, 8].

The supplementary files `check_three_cycle_global.py` and `check_three_cycle_local.py` verify the finite algebraic data, interpolation gaps, spectrum, actual branch derivative and normalized cubic expression with exact arithmetic. The convexity and invariant-set arguments are proved above; an orbit simulation is not used. Formula (82) is a mathematical specification of the potential. The recurrence evaluates its gradient once per step, but no numerical cost bound for computing its defining integral or minimizer is claimed.

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